

TrES-3b

R-0126

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230529 · created 2026-07-16T05:49:39+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460093.7888 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.026
Transit depth (Rp/Rs)^2	3.82 +/- 1.03 [%]
Orbital Inclination (inc)	81.54 +/- 0.69
Airmass coefficient 1 (a1)	0.992 +/- 0.015
Airmass coefficient 2 (a2)	0.009 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#2 - [598, 273]
Optimal Aperture	5.01
Optimal Annulus	11.11
Transit Duration (day)	0.0545 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.026 vs 0.16309 (deviation 1.27σ)
Duration fit vs published	0.0545 d vs 0.0567 d (deviation 0.25σ)
Mid-time O–C	-5.0 min
Depth SNR	7.5
Residual scatter	1.5 %

Light curve

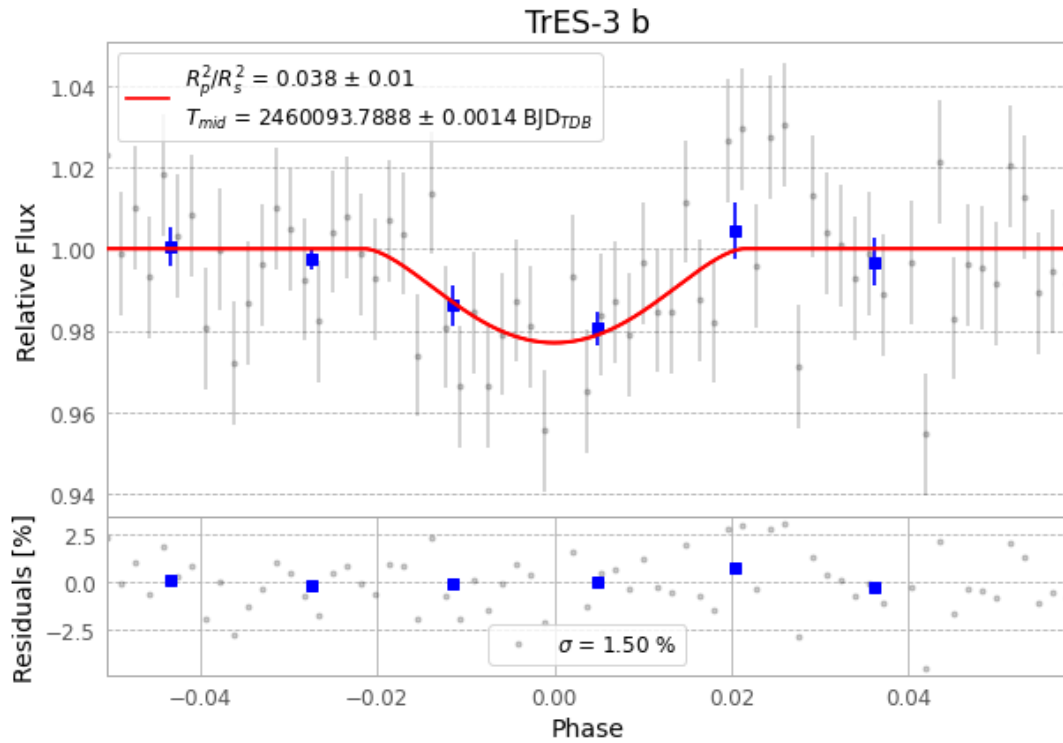


Figure 1 — FinalLightCurve_TrES-3 b_2023-05-28.png (SHA-256 c9f2b3fd5fcfad2b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 56e0a82d4884d798...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

69 FITS frames (45 MB), TRES-3230529051515.FITS → TRES-3230529083915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230529.log (3601541c2ea1...), FinalLightCurve_TrES-3 b_2023-05-28.png (c9f2b3fd5fcf...), FinalLightCurve_TrES-3 b_2023-05-28.csv (6d061de7cd1f...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:49Z.